4.2.1.3 Resistance

We can use an electrical circuit to determine the resistance of a component, as resistance can be calculated from measurements of potential difference and current. We can then see how the resistance is affected by factors such as the length of a wire or when several components are combined.

A dimmer switch is a device that alters the resistance of a circuit and allows you to control the brightness of a lamp. You will investigate the principle of how a dimmer switch works. You will construct a circuit to measure the potential difference across a wire and the current through the wire. You will do this for different lengths of wire and for resistors in series and in parallel.

Learning outcomes	Maths skills required	Formulae
• Set up a circuit to investigate resistance. • Make measurements using ammeters and voltmeters. • Use circuit diagrams to construct and check series and parallel circuits.	• Display experimental data in a suitable table. • Plot two variables from experimental data. • Understand that $y = mx + c$ represents a linear relationship. • Understand and use the symbol $\propto$ • Use an appropriate number of significant figures.	• $V = IR$ • resistance = potential difference ÷ current $R = \frac{V}{I}$

Apparatus list

- d.c. power supply
- voltmeter
- ammeter
- length of resistance wire mounted on a metre rule
- connecting leads
- crocodile clips
- two 10 Ω resistors
- eye protection
- ohmmeter (if available)

Safety notes

- Short lengths of wire are likely to get hot. Use low values of potential difference. Switch off between readings.
- You can add a bulb to your circuit to stop the wire from getting too hot.

Common mistakes

- If your readings keep fluctuating, try to get an average value. Ammeters and voltmeters rarely stay at an exact value.

Methods

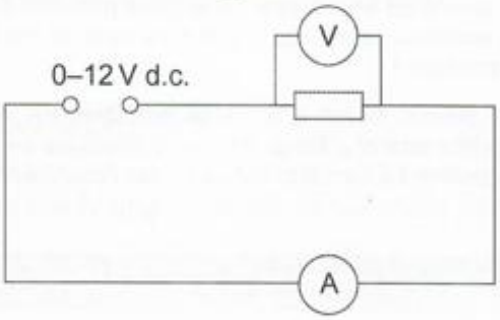
Read these instructions carefully before you start work.

There are two activities in this practical.

Activity 1 – Investigating how the length of a wire affects resistance

1. Set up the circuit as shown in **Figure 1**. The rectangle (symbol for a resistor) is where you should place the component that is being tested.

Figure 1



2. Attach your component, a metre rule with a wire, where the resistor symbol is. Use crocodile clips to add this component in to your circuit as shown in **Figure 2**.

Figure 2



3. Place the crocodile clips so they are 100 cm apart.
4. Turn on the power pack at 4 V.

   Measure both the potential difference across the wire and the current through the wire. Record these results in **Table 1**.
5. **Turn off the power pack**.

   Reduce the length of the wire between the crocodile clips by 10 cm.
6. Repeat steps **4** and **5** for all wire lengths down to 10 cm.

   Be careful – the wire may get hot. Turn off the power pack if the wire starts glowing or smoking.
7. Calculate the resistance for each length of wire using the formula:

   resistance = potential difference ÷ current $\qquad R = \frac{V}{I}$

   Record the values of resistance in **Table 1**.
8. Plot a graph of resistance against length of wire in the space for **Graph 1**:
   - **Resistance in Ω** on the *y*-axis
   - **Length of wire in cm** on the *x*-axis.

Activity 2 – Investigating resistors in series and in parallel

1. Set up the circuit shown in **Figure 1** with two resistors in series where the rectangle is.
2. Turn on the power pack at 4 V.
3. Measure the potential difference across the power pack.
4. Measure the potential difference across each individual resistor and the current through them.
5. Calculate the total resistance of the circuit with resistors in series.
6. Set up the circuit for two resistors in parallel.
7. Repeat steps **2–4** for the resistors in parallel.
8. Calculate the total resistance of the circuit with resistors in parallel.

Record your results

Table 1 – Calculating resistance

Length of wire (cm)	Current (A)	Potential difference (V)	Resistance (Ω)
100			
90			
80			
70			
60			
50			
40			
30			
20			
10			

Graph 1

